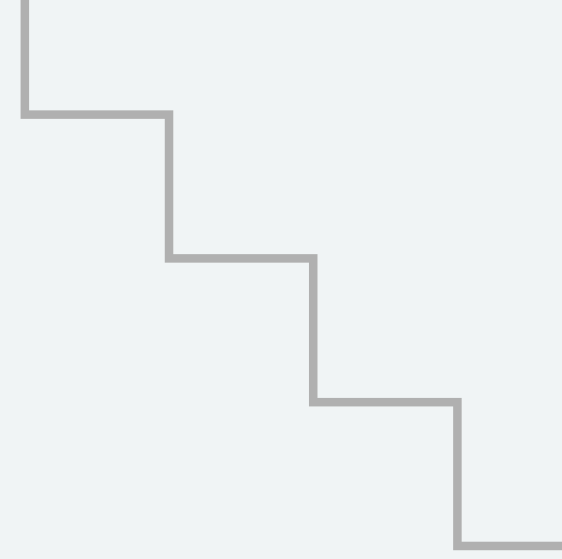




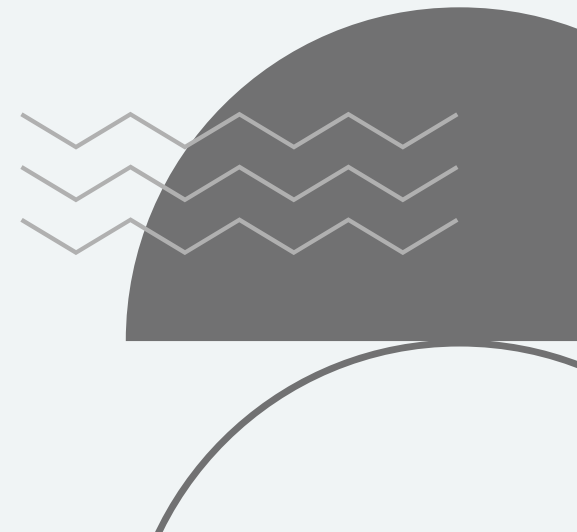
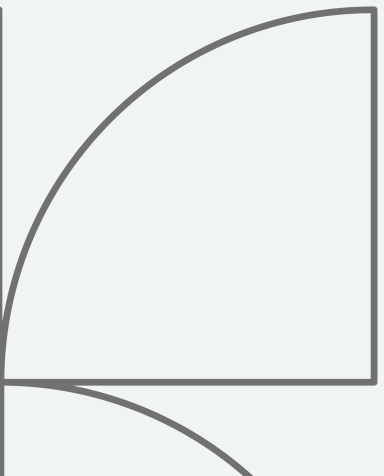
# INTERNSHIP RECOMMENDATION ENGINE

FOR PM INTERNSHIP SCHEME

By,

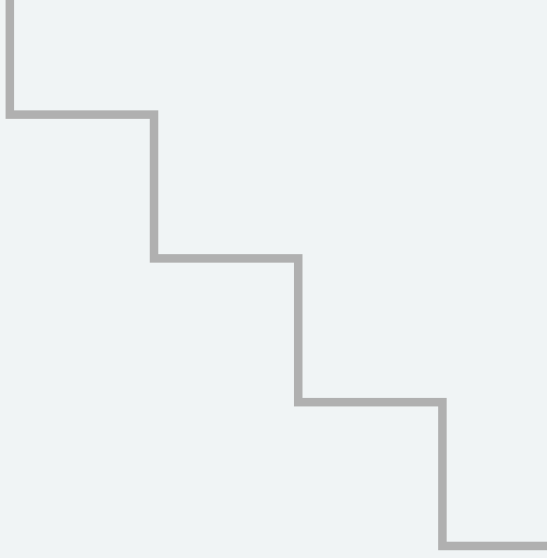
2303717673721020-Krishna Sanjay J

2303717673721022-Madhan G

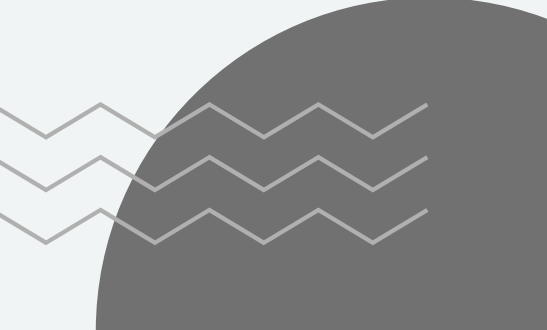
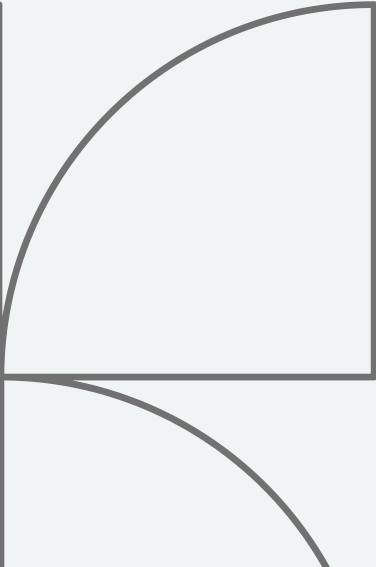




# INTRODUCTION

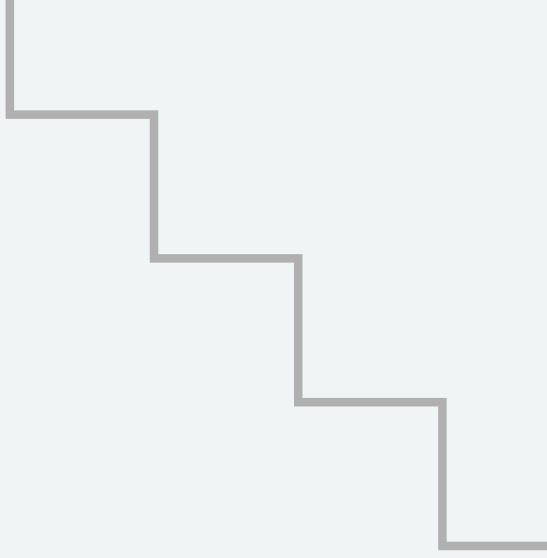


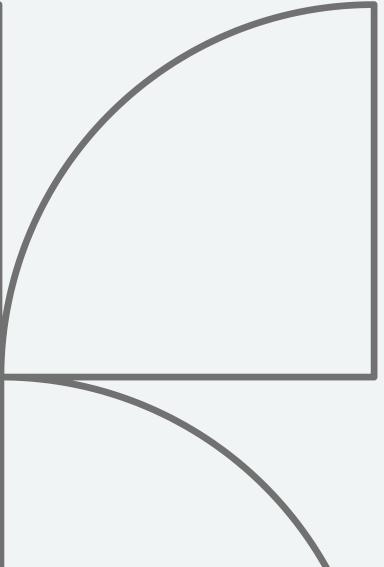
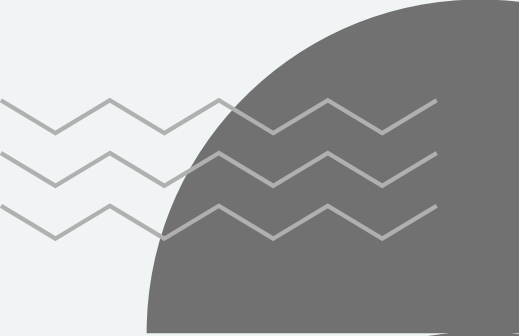
Our project is to finding the right internship under the PM Internship Scheme which is challenging due to large volumes of opportunities. This project builds a Machine Learning and rule-based Recommendation Engine that matches students with suitable internships based on their skills, qualifications, and preferences. Using models like Random Forest and XGBoost, it provides accurate and personalized recommendations.





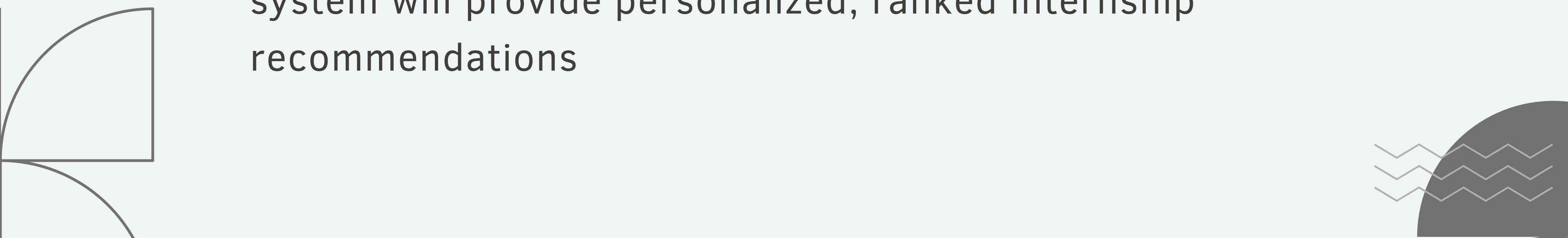
# PROBLEM STATEMENT



- The PM Internship Scheme provides students with a wide range of internship opportunities across government departments and partner organizations.
  - However, with numerous internships available, candidates often face difficulty in identifying positions that best match their skills, education, interests, and career aspirations.
  - Traditional approaches rely on manual search or basic filtering, which is time-consuming, inefficient, and may not capture true candidate-internship compatibility.
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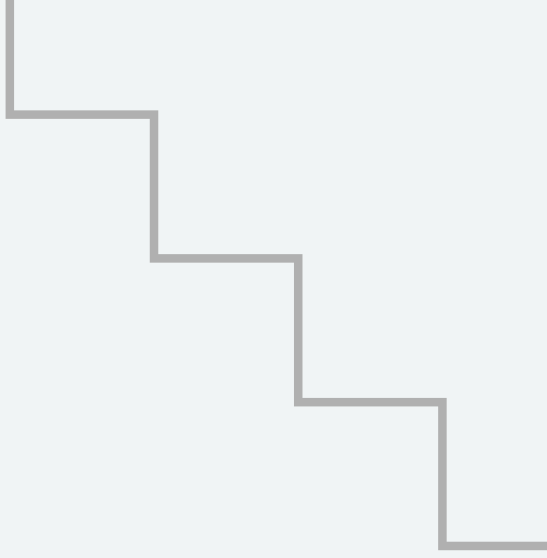


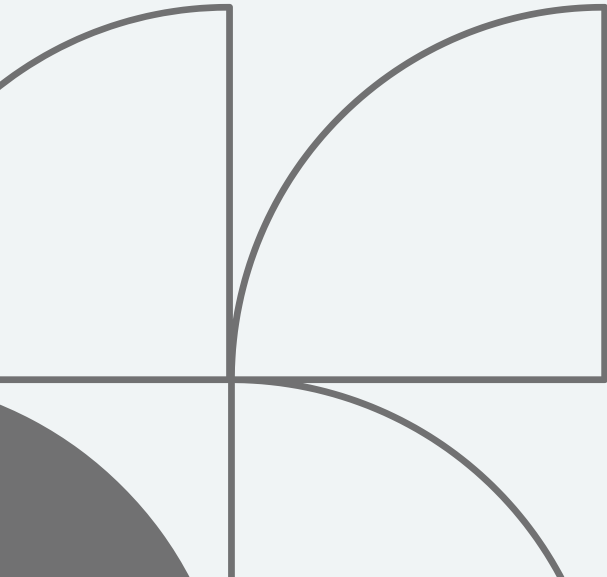
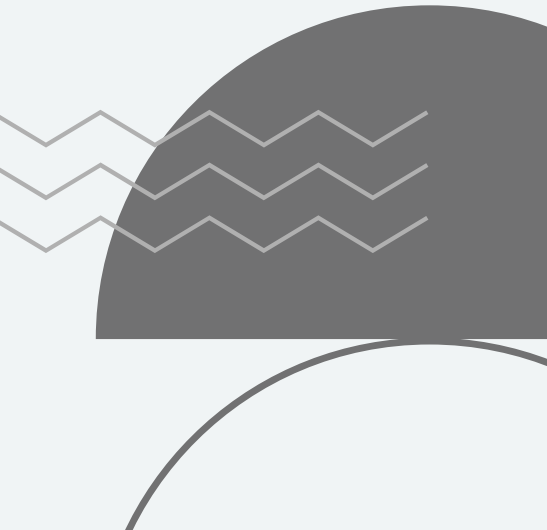
# OBJECTIVE

- The goal of this project, “Internship Recommendation Engine for PM Internship Scheme”, is to develop an intelligent system that automatically predicts the suitability of a candidate for different internships under the scheme.
  - By analyzing candidate profiles (skills, educational background, preferred locations, and sectors) and internship requirements, the system will provide personalized, ranked internship recommendations
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# TECHNOLOGY STACK USED

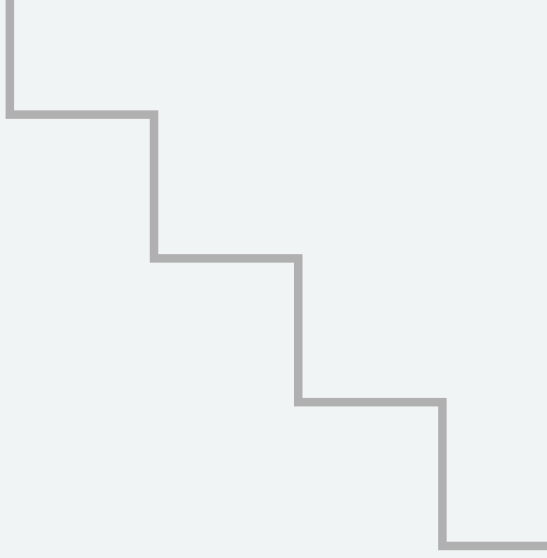


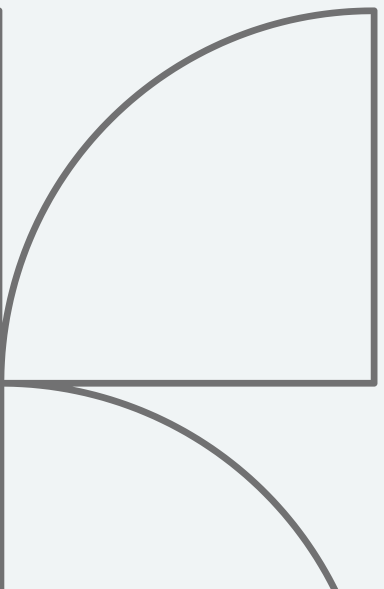
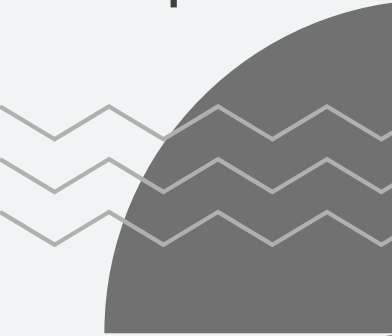
- **Language:** Python
  - **Frontend:** Streamlit (Interactive Web App)
  - **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
  - **Machine Learning:** Scikit-learn, XGBoost, LightGBM
  - **Model Storage:** Pickle (for saving trained models & encoders)
  - **Feature Extraction:** TF-IDF Vectorizer
  - **Tools & Environment:** Jupyter Notebook, VS Code
  - **Dataset:** company.csv (Internship data for training & analysis)
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# METHODOLOGY



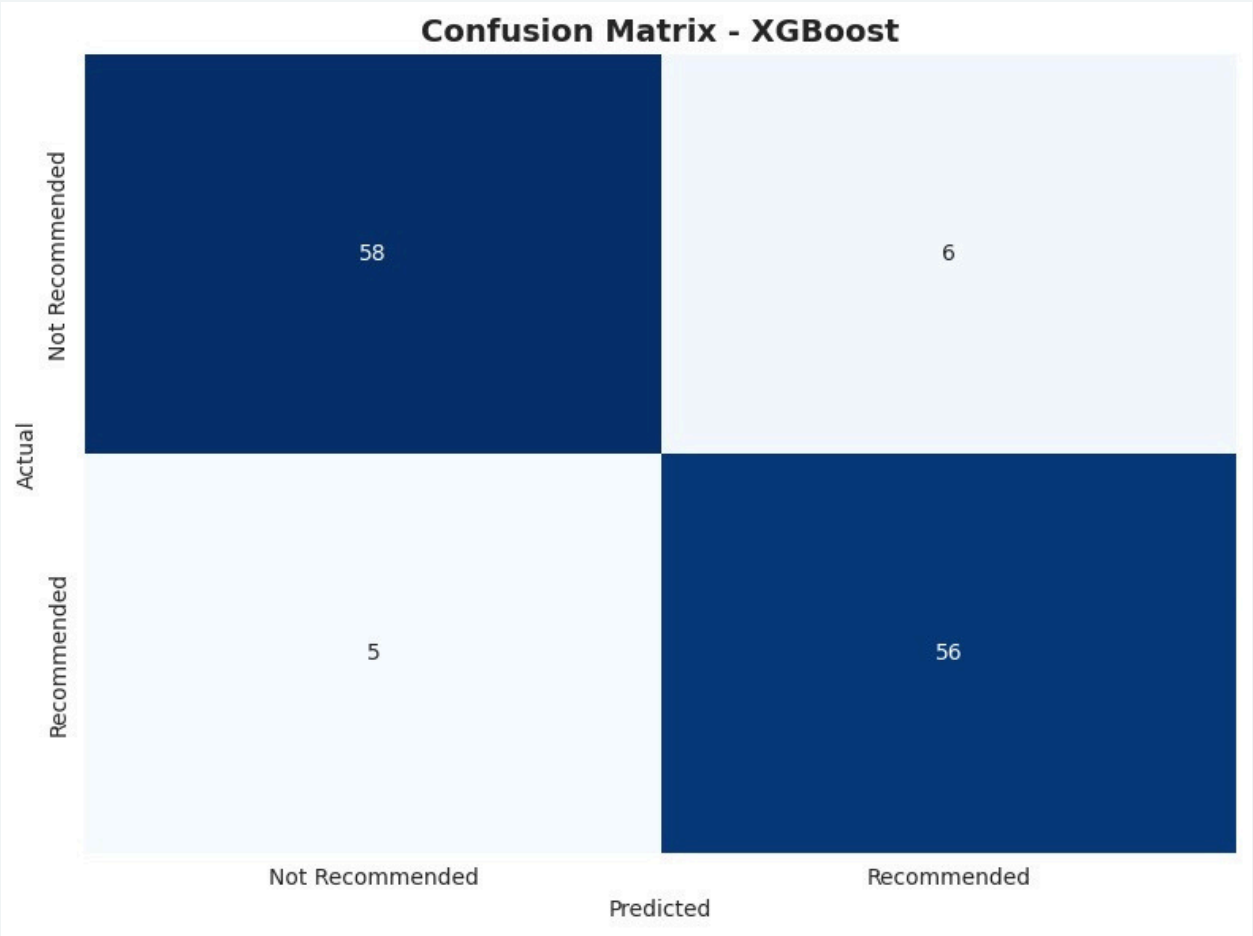
- **Data Collection:** Internship-related data such as company name, role, stipend, location, and required skills were gathered and organized into a dataset.
  - **Data Preprocessing:** Missing values were handled, categorical features were encoded, and numerical features were scaled for uniformity.
  - **Feature Engineering:** New attributes like skill count, degree eligibility, stipend tier, and company size were created to enhance model accuracy.
  - **Model Training:** Multiple machine learning models — including Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM — were trained and tested.
  - **Ensemble Learning:** A Stacking Classifier combining the strengths of different models was used to improve performance and accuracy.
  - **Recommendation Generation:** Based on user inputs such as skills, degree, location, and stipend preference, the trained model ranks and recommends the most suitable internships.
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# RESULT & EVALUATION

## Model evaluation

| Model         | Accuracy | Rank |
|---------------|----------|------|
| XGBoost       | 0.912    | 1    |
| LightGBM      | 0.896    | 2    |
| Decision Tree | 0.856    | 3    |
| Random Forest | 0.816    | 4    |

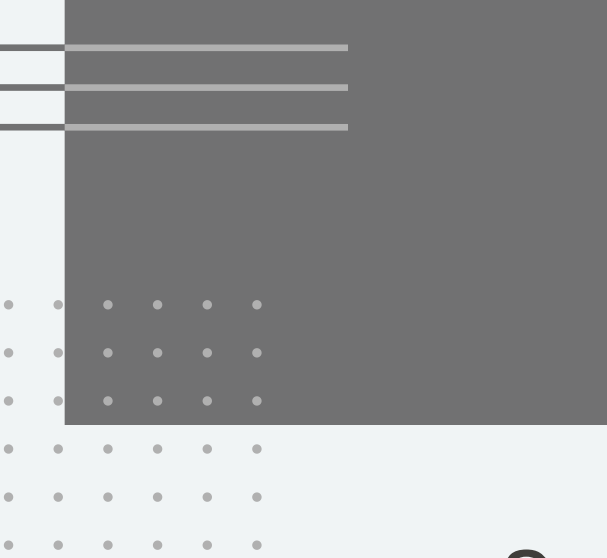
## DETAILED EVALUATION: XGBoost



# RESULT & EVALUATION

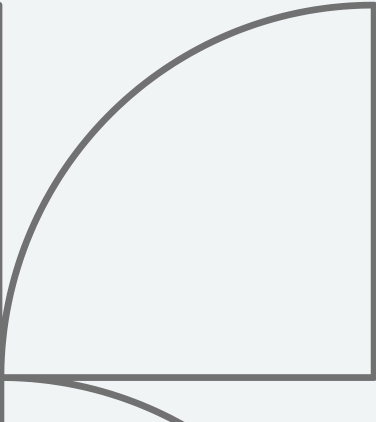
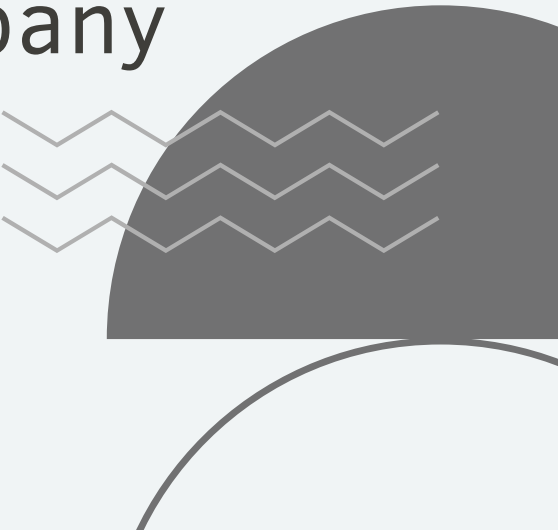


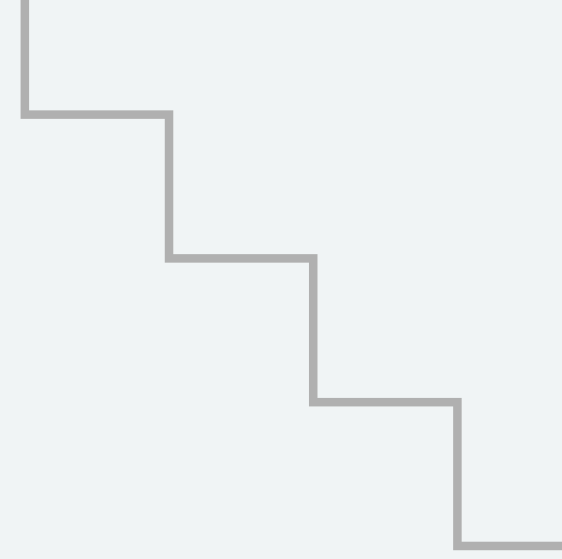




# CONCLUSION



- Successfully developed an internship recommendation system tailored to user skills, degree, location, and preferences.
  - Feature engineering and text-based skill analysis significantly improved recommendation accuracy.
  - The stacking ensemble model provided the best performance among all models tested.
  - Users can get personalized internship suggestions, making the search process faster and more efficient.
  - Future improvements could include real-time data updates, company reviews, and advanced skill matching.
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**THANK  
YOU**

